



ResViT FusionNet Model: An explainable AI-driven approach for automated grading of diabetic retinopathy in retinal images

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ABSTRACT

Background and Objective: Diabetic Retinopathy (DR) is a serious diabetes complication that can cause blindness if not diagnosed in its early stages. Manual diagnosis by ophthalmologists is labor-intensive and time-consuming, particularly in overburdened healthcare systems. This highlights the need for automated, accurate, and personalized machine learning approaches for early DR detection and treatment. Although several deep learning models have been widely used for DR diagnosis, Vision Transformers have recently demonstrated superior image analysis capabilities by capturing long-range dependencies. A hybrid model named ResViT FusionNet has been proposed to improve the accuracy of DR detection in this work.

Methods: For multiclass fundus image classification of diabetic retinopathy, we designed the ResViT FusionNet model, which integrates the robust feature extraction capabilities of Convolutional Neural Networks (CNNs), specifically ResNet50, with the comprehensive understanding provided by Vision Transformers (ViTs). To ensure balanced datasets and enhance model performance, various preprocessing techniques, including data augmentation, were applied. These techniques included rescaling pixel values, horizontal flipping, rotation, and zooming, aimed at improving model generalization and robustness. Additionally, to make the model's predictions more transparent and interpretable, especially in clinical settings, we employed Explainable AI (XAI) techniques. LIME was used to interpret the model's predictions, and Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to generate heatmaps, highlighting the areas of the fundus images that contributed most to the classification decisions. These visual explanations not only enhance trust in the model but also help healthcare professionals understand the underlying factors influencing the predictions.

Results: The experimental findings show that ResViT FusionNet surpasses the performance of leading CNNs and standard ViT models. The model achieved impressive evaluation metrics, including a 0.9307 Precision, 0.9300 Recall, 0.9275 F1 Score, 0.9301 Accuracy, 0.8944 MCC, 0.8935 Kappa and a Jaccard Index of 0.8749.

Conclusions: These findings demonstrate that ResViT FusionNet is a powerful tool for assisting ophthalmologists in making precise, personalized, and timely diagnostic decisions in the evaluation of diabetic retinopathy (DR). The model's integration of CNN and ViT capabilities, along with its explainability through XAI techniques like LIME and Grad-CAM, positions it as an advanced solution for DR detection and classification. By combining the strengths of both convolutional neural networks and vision transformers, and offering transparent insights into its decision-making process, the model enhances diagnostic accuracy and trustworthiness, making it a valuable asset in clinical settings.

1. Introduction

Diabetic retinopathy is a medical condition primarily caused by elevated blood sugar levels, commonly linked to diabetes. High blood sugar can harm the retina, the part of the eye responsible for processing light and sending visual signals to the brain through the optic nerve at the back of the eye [1]. Excess sugar is thought to damage the eyes by blocking the small blood vessels that nourish the retina, leading to fluid

leakage or bleeding [2]. Diabetic retinopathy is typically classified into three stages: normal, non-proliferative DR (NPDR), and proliferative DR (PDR), based on disease progression [3]. NPDR is further classified into mild, moderate, and severe stages, based on the level of disease progression. Patients with NPDR have early symptoms in their retina, such as capillary breaks, dot haemorrhages (HE), and microaneurysms (MA). Severe NPDR is the stage at which the disease advances and small

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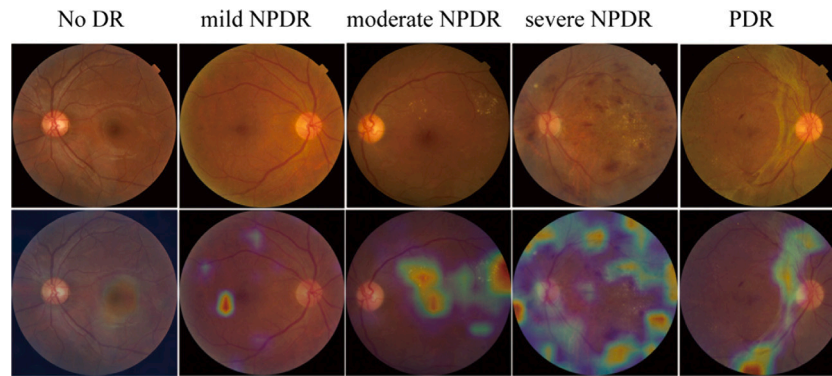


Fig. 1. Fundus images with different severity levels of DR with Grad-CAM highlighting the regions of DR lesions.

blood vessel bulges are discovered. These bulges may eventually begin to leak onto the retina's surface. Severe vision loss results from blood leaking onto the retina's surface in the advanced stage of PDR. The number and type of lesions indicate the disease's severity [4].

In PDR, new blood vessels develop in the retinal area, obstructing the blood supply to the retina. The (Fig. 1) shows the difference of retina from normal to high stages of DR. At initial stages, domain specialists are needed for the tedious, time-consuming process of DR detection and grading [5]. Furthermore, there is a great deal of variation among different doctors based on the visual assessment of patients of DR. Around 79% of DR patients come from developing or underdeveloped countries, which lack access to basic DR detection equipment and ophthalmologists [6]. Manual screening methods are unable to meet the growing demand for diagnostic approaches, given the rising global prevalence of diabetic retinopathy (DR) [7]. While DR detection has demonstrated the robustness of machine learning-based systems, their effectiveness is heavily reliant on handmade features that are challenging to generalize [8]. To get over these limitations, an ophthalmologist may examine any condition caused by DR in real time With the implementation of a computer-aided diagnosis (CAD) system. The identification of many conditions, including diabetic retinopathy, has been greatly aided by deep learning (DL), which has transformed several industries, including medical imaging.

Deep learning has recently introduced transformative methodologies, including Vision Transformers (ViTs), first presented by Dosovitskiy et al. [9], which have demonstrated exceptional results in healthcare. However, while traditional Convolutional Neural Networks (CNNs) excel at extracting localized spatial features, they often struggle to capture long-range dependencies and contextual relationships critical for complex medical imaging tasks like diabetic retinopathy (DR) detection. Conversely, ViTs leverage an attention-based mechanism to assess the importance of various sections of the input sequence [10], but their reliance on large datasets and limited capacity for extracting fine-grained local features can hinder their effectiveness in medical domains with smaller or imbalanced datasets. DL techniques, such as CNNs and hybrid methods, have been widely applied to various medical applications, including diagnosis and treatment recommendations [11, 12]. This research introduces ResViT FusionNet, a hybrid model that addresses these limitations by combining the robust local feature extraction capabilities of CNNs with the global contextual understanding provided by ViTs. The hybrid approach enhances model performance and generalization in DR detection by leveraging the strengths of both architectures. Additionally, the integration of a multilevel classification structure and Explainable AI techniques ensures that the model not only achieves superior diagnostic accuracy but also highlights the critical image regions influencing its predictions. By bridging the gaps in existing models, ResViT FusionNet aims to provide a more reliable, interpretable, and clinically applicable solution for automated DR diagnosis. The key contributions outlined in this paper can be summarized as follows:

- We propose a hybrid model called ResViT FusionNet, which combines Convolutional Neural Networks (CNNs) with Vision Transformers (ViTs), integrating the strengths of both architectures for improved diabetic retinopathy (DR) diagnosis.
- The CNN component, ResNet50, excels in robust feature extraction, while the ViT component enhances holistic understanding by capturing long-range dependencies in retinal images.
- The proposed ResViT FusionNet model performs multilevel classification of diabetic retinopathy (DR) using the APTOS 2019 dataset, which aids ophthalmologists in more effectively examining and diagnosing DR.
- Explainable AI (XAI) techniques, including LIME and Grad-CAM, are incorporated to enhance the transparency and interpretability of the model's predictions, ensuring clinical reliability and trustworthiness.
- The effectiveness of the ResViT FusionNet model is evaluated by comparing its Accuracy, F1-score, Matthew's Correlation Coefficient, Cohen's Kappa, and Jaccard Index Score to those of existing state-of-the-art ML methods.

The paper is structured as follows: Section 2 examines the epidemiological trends in diabetic retinopathy (DR), while Section 3 is about related literature. Sections 4 and 5 outlines the proposed framework for multilevel DR classification and emphasizes key findings. Section 6 assesses the model's performance in comparison to state-of-the-art techniques, and Section 7 presents the conclusion. Finally, Section 8 discusses potential future research directions.

2. Epidemiological trends in diabetic retinopathy

Diabetic Retinopathy (DR) represents a significant and multifaceted challenge within the realm of diabetes complications. The intricate nature of DR is exacerbated by its asymptomatic early stages, where subtle retinal changes occur long before any noticeable vision impairment. This silent progression complicates early detection and intervention, often resulting in patients presenting with advanced disease stages where irreversible damage has already occurred. Furthermore, the systemic nature of diabetes contributes to DR's severity, as chronic hyperglycemia not only accelerates retinal vascular damage but also impairs the retina's ability to heal and regenerate. The International Diabetes Federation (IDF) published the most recent edition of the Diabetes Atlas, which shows that there were approximately 537 million diabetics globally in 2021 and this figure is projected to rise to 783 million by 2045 [13]. According to projections made by the International Agency for the Prevention of Blindness (IAPB), 224 million individuals globally, would be affected by diabetic retinopathy by 2040 (Fig. 2). Of these, 70 million individuals (11%) are affected by diabetic retinopathy, which poses a significant risk to vision [14]. Consequently, DR, a severe microvascular complication of diabetes, significantly impacts global health systems.

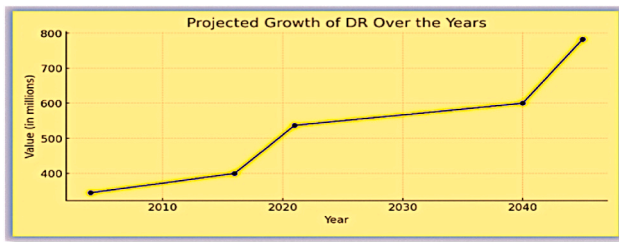


Fig. 2. Growth Rate of DR Patients over the years.

3. Related work

Methods of Deep learning have made significant strides in the health domain, establishing itself as a powerful tool across various medical fields. Its advancements have led to the evolution of numerous deep learning architectures. They were successfully applied in areas such as disease segmentation [15,16] and disease prediction [17,18]. Among these, the Vision Transformer (ViT), a deep learning-based architecture, has shown remarkable results in the healthcare sector. Fan et al. [19] developed a DR classification approach based on multiscale feature fusion with an adaptive weighted operation. This work achieved 85.32% of accuracy using APTOS dataset. Sugeno et al. [20] developed a system for grading diabetic retinopathy (DR) severity using transfer learning with the EfficientNet-B3 architecture. They achieved 84.2% accuracy for severity grading tasks using APTOS dataset. Nneji et al. [21] introduced a method for DR classification based on a capsule network, employing contrast-limited adaptive histogram equalization as a data preparation step for retina fundus images and utilizing the capsule network for feature extraction. Their approach, which used a sigmoid classifier, reached 99.1% accuracy on the Messidor dataset. Bodapati et al. [22] introduced a composite deep neural network (DNN) integrated with a gated-attention mechanism for DR classification, achieving 97% Kappa score and 82.54% accuracy. Additionally, they developed a DR detection method that employed a stacked convolutional autoencoder with spatial attention [23], resulting in 84.17% and 63.24% accuracies on the APTOS dataset and IDRid dataset respectively. Shaik and Cherukuri [24] introduced the Hinge Attention Networks (HA-Net) model for classifying DR severity levels, achieving an accuracy of 85.54% on the APTOS dataset and 66.41% using the IDRid dataset. Kumar et al. [25] presented a novel architecture called DRISTI, a hybrid deep learning model that combines VGG16 with a capsule network. This model achieved 82.06% accuracy for five-class classification and 96.24% accuracy for binary classification. Jabbar et al. [26] developed a DR diagnosis system using a VGG model based on transfer learning. They achieved 96.6% accuracy using the Kaggle EyePACS dataset. Additionally, Li et al. [27] introduced a cross-disease attention network named CANet for diagnosing DR and diabetic macular edema. Their model used individual attention modules for feature selection, which were integrated into a deep network. This method achieved 92.6% accuracy using Messidor dataset and 65.1% accuracy using the IDRid dataset.

Al-Antary and Arafa [28] introduced a multi-scale attention network for detecting diabetic retinopathy. Their method utilizes an attention mechanism to extract features from retinal images and was tested on the APTOS and EyePACS datasets. The network achieved accuracies of 98.1% and 87.5% for DR detection, and 84.6% and 79.9% for DR severity classification, respectively. Mohan et al. [29] presented a Vision Transformer-based model called ViT-DR for detecting diabetic retinopathy severity. Their approach involves dividing fundus images into non-overlapping patches, which are then converted into sequences for processing, and then processing these sequences through positional embeddings and multiple multi-head attention layers. The model was

tested on both the Kaggle and IDRid datasets. R. Sun et al. [30] introduced the Lesion Aware Transformer (LAT), incorporating an encoder to capture pixel relationships and a decoder specifically designed for lesion-aware localization. The LAT model approaches lesion detection as a weakly supervised localization problem and achieved state-of-the-art results on several datasets. Yu et al. [31] developed the MIL-ViT (Multiple Instance Learning-Vision Transformer) model, pretrained and then fine-tuned for retinal disease classification tasks. The model leverages MIL-based headers in a plug-and-play fashion with Vision Transformers. Experiments on the APTOS 2019 and RFMid2020 (Retinal Fundus Multidisease Image Dataset) demonstrated that MIL-ViT outperforms CNN-based baseline models. Yang et al. [32] introduced a hybrid model that combines convolutional and transformer layers to classify fundus diseases using the OIA (Ophthalmic Images Analysis) datasets. In [33], the authors presented a hybrid architecture that integrates a Vision Transformer with a Support Vector Machine (ViT-SVM) for grading of Optical Coherence Tomography (OCT) scans, with the goal of automating the early diagnosis of ocular disorders. Following training, the VGG-16 model got 83% accuracy, while the ViT model got 88% accuracy. The proposed method surpassed all other models with 94% of accuracy.

This paper [34] proposes a deep learning-based multi-label classification model using Grad-CAM for diabetic retinopathy fundus images. The model not only classifies DR but also identifies and locates various lesions, reducing the need for extensive manual annotation. Five lesion labels were pre-defined, and 3228 fundus images were used for model development. The model, built on a custom ResNet architecture, achieved a sensitivity of 93.9% and specificity of 94.4% in DR classification, while accurately outlining lesion regions in the images. In this study [35], authors propose a lightweight custom convolutional neural network (CNN) for DR diagnosis using optical coherence tomography (OCT) images, with potential applications in IoT environments. Using pre-trained MobileNet for transfer learning, they fine-tuned the model for robust OCT classification. Then employed Grad-CAM to highlight critical regions in the images. The model, evaluated on metrics such as accuracy, precision, recall, and ROC Area, achieved 80% accuracy, 85% precision, and 80.5% recall. Artificial intelligence (AI) can assist doctors by reducing diagnostic time [36]. While state-of-the-art convolutional neural networks (CNNs) for DR detection achieve 75% accuracy, they are black-box models, making their decisions hard to interpret. This lack of interpretability can hinder trust in the model's output, highlighting the need for explainable AI techniques. In this study, authors enhanced the CNN-based model, achieving 92% accuracy, and apply Grad-CAM to provide visual explanations of the regions the model considers significant. This approach presents both classification results and their explanations, helping medical practitioners make informed decisions. This paper [37] explores the use of ensemble learning methods to enhance the prediction accuracy for DR detection from retinal images, alongside the application of Grad-CAM to improve model interpretability. Using a dataset of 1437 color fundus images, they evaluated several pre-trained convolutional neural networks (CNNs)—Xception, VGG16, InceptionV3, and DenseNet121—with accuracies of 89.27%, 91.44%, 89.06%, and 93.35%, respectively. To further enhance accuracy, they employed weighted voting, achieving 93.95%, and meta-learning, which improved the accuracy to 94.63%. These methods outperformed individual models in distinguishing between non-proliferative and proliferative DR phases. Additionally, Grad-CAM provides visual explanations, improving interpretability and supporting the development of reliable diagnostic tools for early DR detection and clinical use.

The study [38] proposed an automated method to identify diabetic retinopathy from ocular angiography using convolutional neural networks (CNNs). Two models are presented: the first discriminates between healthy eyes and those with retinopathy, while the second distinguishes between non-proliferative retinopathy and weakly or severely proliferative retinopathy. The models achieved accuracies of 0.98 and

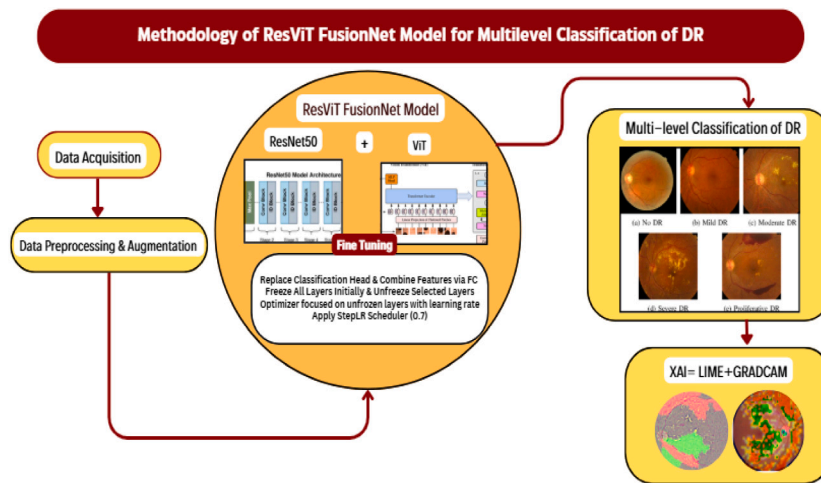


Fig. 3. Block diagram of the methodology for Diabetic Retinopathy multi-stage classification with Explainable AI integration.

0.91, respectively, demonstrating their effectiveness in aiding diagnosis. Additionally, the method localizes the disease in angiography, providing explainability through class activation mapping (CAM) algorithms that highlight areas indicative of the disease. This enhances model trustworthiness for both clinicians and patients. They also introduce a similarity index to evaluate model robustness by quantifying the differences in heatmaps generated by the CAM algorithms for the same model. This study [39] introduces an XAI-enabled teleophthalmology model, XAITO-DRGC, designed for grading and classifying diabetic retinopathy (DR) in remote settings. By leveraging OphthoAI IoMT headsets, the model enables continuous DR monitoring and applies techniques like median filtering, contrast enhancement, U-Net-based segmentation, and SqueezeNet feature extraction. The XAITO-DRGC employs the Archimedes optimization algorithm (AOA) and a bidirectional gated recurrent convolutional unit (BGRUCU) for DR detection and classification. Experimental results demonstrate that XAITO-DRGC outperforms existing models across multiple benchmarks. The study [40] presents OptiDex, a deep learning-based model for detecting and classifying diabetic retinopathy (DR) severity. It incorporates Explainable AI (XAI) for transparent decision-making and uses an enhanced cat swarm optimization (ECSSO) algorithm to improve performance. OptiDex employs morphological operations, hysteresis thresholding, and the Nas-Mob architecture to segment and extract features from retinal images. Evaluated on clinical and public datasets (Aptos, IDRiD, Messidor), OptiDex achieved high accuracy (97.65%), sensitivity (96.46%), and specificity (93.45%) in detecting DR and identifying its severity levels. This paper [41] introduces ExplAI, an eXplanatory AI (XAI) model for classifying diabetic retinopathy (DR) severity using Color Fundus Photography (CFP). Unlike typical “black-box” AI models, ExplAI segments and categorizes lesions autonomously, deriving image-level DR classifications from these segmentations while maintaining transparency. The model is trained end-to-end, with lesion localization achieved through self-supervision, transforming occluded areas into healthy-looking regions for clarity. Evaluated on multiple CFP datasets, ExplAI combines high classification accuracy with interpretability, potentially enhancing AI adoption in medical settings. This paper [42] presents a computer-aided detection system for diabetic retinopathy (DR), utilizing retinal fundus images to identify and classify lesions indicative of DR. The system aids ophthalmologists by automatically categorizing retinas as healthy or unhealthy and offers an interactive tool for doctors to review highlighted suspicious lesions in unhealthy cases. Built on a Faster R-CNN ResNet 101 architecture, the system achieves high sensitivity and specificity on benchmark datasets Messidor-2, IDRiD, and E-Ophtha. This tool aims to support timely DR diagnosis and enhance clinical decision-making, leaving the final diagnosis to eye specialists.

Apart from medical image classification, Vision Transformers (ViTs) are widely used in various other medical imaging tasks such as medical image segmentation [43], medical image detection [44], medical image reconstruction [45], medical image synthesis [46], and medical image report generation [47]. Convolutional Neural Networks (CNNs) have demonstrated significant advantages in medical imaging as they automatically and hierarchically learn spatial features from images. They excel in capturing local patterns and textures through their convolutional layers, making them constructive for jobs like detection of object and to classify the images. The combination of Vision Transformers and CNNs leverages the strengths of both architectures. CNNs are effective at extracting fine-grained features and learning spatial hierarchies from local image patches, while ViTs excel in capturing long-range dependencies and global context through their attention mechanisms. Integrating CNNs with ViTs can enhance feature representation by combining detailed local information with a broader contextual understanding, leading to improved performance in medical imaging tasks and more robust and accurate diagnostic tools. We proposed a hybrid model combining Vision Transformers and Convolutional Neural Networks. Additionally, we employed LIME to interpret the model's predictions, and Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to generate heatmaps, highlighting the areas of the fundus images that contributed most to the classification decisions (Fig. 3).

4. Proposed ResViT FusionNet model for multi-level classification of DR

We aim to integrate the Convolutional Neural Network (CNN), specifically ResNet50, with a Vision Transformer (vit_base_patch32_224) for the multilevel classification of diabetic retinopathy (DR).

4.1. ResNet50 model

ResNet50 is a powerful image classification model (Fig. 4) recognized for its ability to achieve state-of-the-art performance on extensive datasets. The main innovation of ResNet50 lies in its implementation of residual connections, which significantly improve the model's ability to train deeper architectures without the vanishing gradient problem.

4.1.1. Residual connections

ResNet50 utilizes residual connections that enable the network to understand residual functions, which map the input to the output. This method aids in training very deep networks by reducing the vanishing gradient problem.

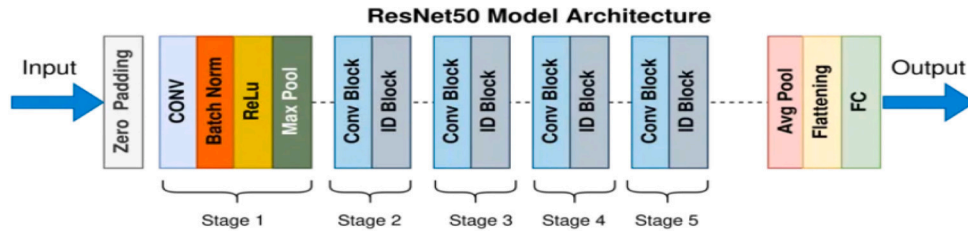


Fig. 4. ResNet50 architecture.

4.1.2. Architecture components

The ResNet50 architecture is comprised of the following four primary components:

- **Convolutional Layers:**
 - Extract features from the image in input layer, including edges, shapes and textures.
 - Comprise multiple convolutional layers, each followed by batch normalization and ReLU activation.
 - Incorporate max pooling layers to decrease the feature map's spatial dimensions while preserving important features.
- **Identity Block:**
 - Processes the input through multiple convolutional layers.
 - Reinforces the output by adding the input back, which facilitates the learning of residual functions.
- **Convolutional Block:**
 - Same as identity block, but incorporates a 1×1 convolutional layer to decrease the number of filters before applying the 3×3 convolutional layer.
- **Fully Connected Layers:**
 - Utilized for final classification tasks.
 - The output from the last fully connected layer is passed into a softmax activation function to generate class probabilities.

4.1.3. Training and performance

- **Training Dataset:** ResNet50 was trained on the ImageNet dataset, which comprises more than 14 million images spanning 1000 different classes.
- **Performance:** ResNet50 with 22.85% error rate on the ImageNet dataset, which is competitive compared to the typical human error rate of 5.1%.

4.2. Vision transformer model

The Vision Transformer (ViT) modifies the transformer architecture, initially designed for natural language processing (NLP), to suit image recognition tasks (Fig. 5). Transformers are able to trap long-range dependencies in text, are now applied to computer vision, potentially replacing CNNs. For the multiple grading classification of DR, we utilized the fine-tuned vit_base_patch32_224 variant of ViT.

4.2.1. Model configuration

The vit_base_patch32_224 variant has specific settings:

- **Base Model:** Uses the standard configuration of ViT with a moderate number of parameters and layers.
- **Patch Size (32):** Divides the image into 32×32 pixel patches, analogous to processing text sequences in NLP.
- **Input Size (224):** Expects images of 224×224 pixels, balancing detail and computational efficiency.

4.2.2. Patch embedding

- **Image Splitting:** The 224×224 pixel image is divided into 32×32 pixel patches, resulting in 49 patches.
- **Flattening:** Each 32×32 patch is flattened into a 1D vector with a size of 3072 ($32 \times 32 \times 3$).
- **Linear Projection:** Flattened vectors are projected into a higher-dimensional space using a learnable embedding matrix E , mapping to a d -dimensional space ($E : 3072 \times d$).

4.2.3. Patch embedding matrix E

The matrix E is learned during training and transforms each flattened patch vector into a d -dimensional embedding.

4.2.4. Positional encoding

To retain spatial information, positional encodings are added to patch embeddings. These learnable vectors provide information about the relative positions of the patches.

4.2.5. Classification token

A learnable classification token is added to the patch embeddings, which gathers information from all patches and is utilized for the final classification.

4.2.6. Transformer encoder

The sequence of patch embeddings, classification token, and positional encodings is processed through the transformer encoder, consisting of L layers with multi-head self-attention (MSA) and a multilayer perceptron (MLP) block:

• Multi-Head Self-Attention (MSA):

- **Self-Attention:** Computes weighted sums of values based on similarity between queries and keys.
- **Multi-Head Attention:** Utilizes several attention heads to detect the diverse patterns. The attention calculation (equation i) is:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

– Multilayer Perceptron (MLP):

- * **Structure:** Consists of two fully connected layers with a GeLU activation function. The first layer expands the dimensionality, and the second layer reduces it.
- * **Normalization and Residuals:** Layer normalization is applied before MSA and MLP blocks, with residual connections added after each block.

4.2.7. Final classification

The output corresponding to the classification token is used for classification. The output is fed into a linear layer to generate the final class probabilities.

4.2.8. Model characteristics

The (Table 1) represents the characteristic of vision transformer model.

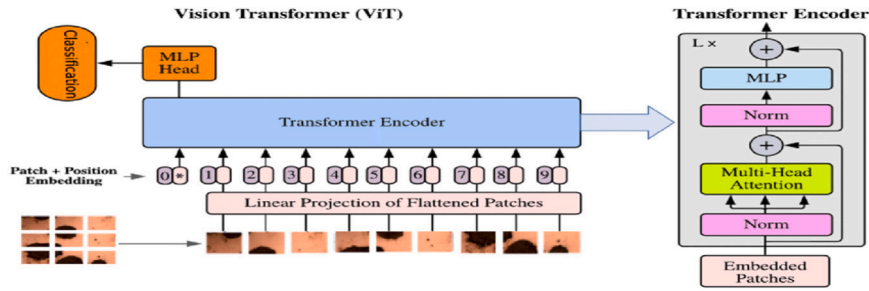


Fig. 5. Vision Transformer model.

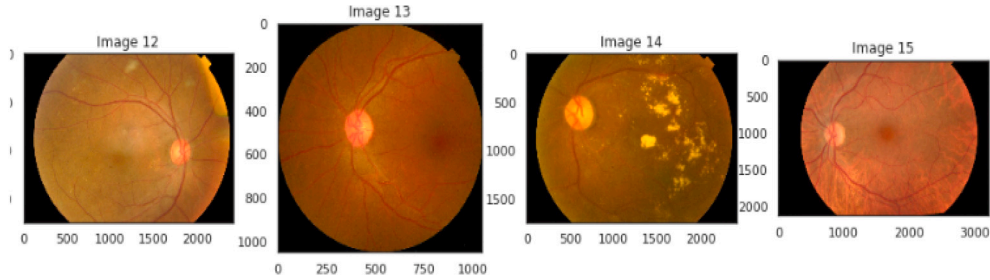


Fig. 6. Image samples of the APTOS 2019 dataset.



Fig. 7. Sample images after Preprocessing steps from the dataset.

Table 1

ViT model characteristics.

Patch size	32×32
Input image size	224×224
Embedding dimension	Typically $d=768$ for the base model.
Total layers	$L=12$ transformer layers.
Total attention heads	Typically 12 heads.
Number of parameters	Approximately 86 million parameters.

5. Experimental setup

5.1. Dataset description

The dataset (Fig. 6) used in this study is the APTOS 2019 dataset, a well-known benchmark dataset for diabetic retinopathy classification [48]. This dataset contains retinal fundus images labeled into five distinct classes, representing different stages of diabetic retinopathy: Class 0 (No DR - Healthy), Class 1 (Mild DR), Class 2 (Moderate DR), Class 3 (Severe DR), and Class 4 (Proliferative DR). The dataset comprises a total of 5590 images, which are split into a Training Set of 3662 images and a Test Set of 1928 images. The training set is further divided into training and validation subsets, with 80% of the images used for training and 20% reserved for validation, ensuring that the model is evaluated on unseen data during the development phase.

5.2. Data preprocessing and augmentation steps

To enhance the generalization capability of the ResViT FusionNet model, various preprocessing techniques were employed, including horizontal flipping, rotation, zooming, and rescaling pixel values (Fig. 7). These techniques were chosen because they simulate common variations in medical imaging data, such as differences in orientation, scale, and perspective, which are frequently encountered in clinical practice. Horizontal flipping ensures that the model remains robust to variations in the positioning of anatomical features, a common occurrence in fundus images due to differences in imaging protocols. Rotation and zooming further increase the diversity of the training set by introducing slight variations that mimic real-world acquisition inconsistencies. Such transformations have been shown to prevent overfitting and improve generalization in medical imaging tasks.

Sunkari et al. [49] demonstrated that augmentation techniques like zooming, rotation, and resizing significantly enhance the performance of deep learning models in the classification of retinal diseases by increasing the diversity of training data. Similarly, Ashwini and Dash [50] applied rotation at different angles along with horizontal and vertical flipping to increase the data samples. Kommaraju and Anbarasi [51] addressed the challenges of data imbalance by employing zooming, shearing, rotation, flipping, and rescaling. Mutawa et al. [52] applied rotation, shifting, flipping, and zooming to complement the model's training data. By employing these proven methods, we ensured that the ResViT FusionNet model is better equipped to handle unseen data in clinical settings.

5.2.1. Rescaling (normalization)

To normalize the data and ensure that the model does not become biased by large pixel values, each pixel value in the image is rescaled to the range [0, 1]. This is achieved by dividing the original pixel value by 255, as shown in the following (equation ii):

$$I_{\text{rescaled}}(x, y, c) = \frac{I_{\text{original}}(x, y, c)}{255} \quad (2)$$

Here:

- $I_{\text{original}}(x, y, c)$ is the original intensity value of the pixel at coordinates (x, y) in the color channel $c \in \{R, G, B\}$, where the values range between 0 and 255.
- $I_{\text{rescaled}}(x, y, c)$ is the normalized pixel value at the same coordinates after rescaling, which lies in the range [0, 1].

This pixel-wise normalization ensures consistency across all pixel values in the image and aids in faster model convergence during training.

5.2.2. Horizontal flip

A random horizontal flip is applied to the images to make the model invariant to left-right orientation. The transformation is performed at the pixel level, as represented in the following (equation iii):

$$I_{\text{flip}}(x, y, c) = I(W - x + 1, y, c) \quad (3)$$

Here:

- $I(x, y, c)$ is the intensity value of the original image at pixel coordinates (x, y) in the color channel $c \in \{R, G, B\}$.
- $I_{\text{flip}}(x, y, c)$ is the intensity value of the horizontally flipped image at the same pixel coordinates.
- W is the width of the image in pixels.

5.2.3. Rotation

Random rotations are applied to the images by rotating each pixel's coordinates by a random angle θ . The transformation is given by (equation iv):

$$I_{\text{rotated}}(x', y', c) = I(x, y, c) \quad (4)$$

where:

- (x', y') are the new pixel coordinates after rotation, calculated as (equation v and vi):

$$x' = \cos(\theta) \cdot (x - x_c) - \sin(\theta) \cdot (y - y_c) + x_c \quad (5)$$

$$y' = \sin(\theta) \cdot (x - x_c) + \cos(\theta) \cdot (y - y_c) + y_c \quad (6)$$

- (x_c, y_c) are the coordinates of the center of the image.
- θ is a randomly selected angle.
- $I_{\text{rotated}}(x', y', c)$ is the intensity value of the rotated image at the new coordinates.

5.2.4. Zooming

Zooming randomly changes the scale of the image. This transformation can be described at the pixel level as (equation vii):

$$I_{\text{zoomed}}(x', y', c) = I(x, y, c) \quad (7)$$

where:

- (x', y') are the new coordinates of the pixel after applying the zoom factor α (equation viii and ix):

$$x' = \alpha \cdot (x - x_c) + x_c \quad (8)$$

$$y' = \alpha \cdot (y - y_c) + y_c \quad (9)$$

- (x_c, y_c) are the coordinates of the center of the image.

- α is the zoom factor, with $\alpha > 1$ for zooming in and $0 < \alpha < 1$ for zooming out.
- $I(x, y, c)$ is the intensity value of the original image at pixel (x, y) in the color channel $c \in \{R, G, B\}$.
- $I_{\text{zoomed}}(x', y', c)$ is the intensity value of the zoomed image.

5.2.5. Width/height shift

Random translations are applied to the image by shifting pixel positions in the x and y directions. The transformation is given by (equation x):

$$I_{\text{shifted}}(x', y', c) = I(x, y, c) \quad (10)$$

where:

- (x', y') are the new pixel coordinates after shifting (equation xi and xii) :

$$x' = x + \Delta x \quad (11)$$

$$y' = y + \Delta y \quad (12)$$

- Δx and Δy are the random shifts in the x and y directions, respectively.
- $I(x, y, c)$ is the intensity value of the original image at pixel (x, y) in the color channel $c \in \{R, G, B\}$.
- $I_{\text{shifted}}(x', y', c)$ is the intensity value of the shifted image.

5.3. Test set preparation

The test set images are not augmented but are rescaled in the same manner as the training images to ensure consistent input to the model. The rescaling is performed at the pixel level as follows (equation xiii):

$$I_{\text{rescaled}}(x, y, c) = \frac{I_{\text{original}}(x, y, c)}{255} \quad (13)$$

Here:

- $I_{\text{original}}(x, y, c)$ is the original intensity value of the pixel at coordinates (x, y) in the color channel $c \in \{R, G, B\}$.
- $I_{\text{rescaled}}(x, y, c)$ is the rescaled intensity value of the pixel, which lies in the range [0, 1].

This ensures that test evaluation is fair and consistent with the training data normalization.

The preprocessing steps ensure that the model is trained on a normalized, diverse, and consistent dataset, with no data leakage between the training, validation, and test sets.

5.4. Fine tuning of ResViT FusionNet model for DR grading

5.4.1. ResViT FusionNet model initialization

All layers of the ResNet50 and Vision Transformer (ViT) are loaded and initialized with pretrained weights. The classification head of ResNet50 is replaced, and a combined fully connected layer is added to fuse features from both networks.

5.4.2. Freezing layers

All layers of ResNet50 and ViT are frozen to preserve the pretrained weights Prevents updates during the initial training phase.

5.4.3. Unfreezing layers

Unfreezes specific layers of the model to allow fine-tuning on new data. The final layer of ResNet50 (layer4), the last 10 transformer blocks of the Vision Transformer (ViT), and the combined classification head (fc_combined) are unfrozen to allow updates.

```

class HybridViTResNet(nn.Module):
def __init__(self, num_classes):
super(HybridViTResNet, self).__init__()
# Load ResNet50 and Vision Transformer models
self.resnet50 = models.resnet50(pretrained=True)
self.resnet50.fc = nn.Identity()
self.vit = create_model('vit_base_patch32_224',
pretrained=True, num_classes=num_classes)
resnet_features = self.resnet50.fc.in_features
vit_in_features = self.vit.head.in_features
self.fc_combined = nn.Linear(resnet_features +
vit_in_features, num_classes)
print('ResNet50 and ViT initialized, fully connected layer
combined.')

```

```

for param in model.resnet50.parameters():
param.requires_grad = False
for param in model.vit.parameters():
param.requires_grad = False
print('ResNet50 and ViT layers frozen.')

```

```

for param in model.resnet50.layer4.parameters():
param.requires_grad = True
for param in model.vit.blocks[-10:].parameters():
param.requires_grad = True
for param in model.fc_combined.parameters():
param.requires_grad = True

```

5.4.4. Setting up loss and optimizer

The loss function and optimizer are configured for fine-tuning. CrossEntropyLoss is used for multi-class classification. The Adam optimizer is used on the unfrozen layers with a learning rate (lr) of 5×10^{-4} .

```

criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(filter(lambda p: p.requires_grad,
model.parameters()), lr=5e-4)

```

5.4.5. Learning rate scheduler

A learning rate scheduler is implemented to adjust the learning rate throughout training. The StepLR scheduler reduces the learning rate by a factor of 0.7 after each epoch.

```

scheduler = StepLR(optimizer, step_size=1, gamma=0.7)

```

By following these steps, the model can benefit from the extensive knowledge gained from large-scale datasets while adapting to perform effectively on specific tasks, even with potentially limited new data. Fine-tuning thus strikes a balance between leveraging pre-existing knowledge and acquiring new patterns tailored to the target dataset.

5.5. Results

This section presents the performance metrics of ResViT FusionNet model to classify DR into various categories: Healthy, Mild DR, Moderate DR, Severe DR, and Proliferative DR. Let us discuss each metric and the results:

5.5.1. Precision

Precision (as defined in equation xiv) is the ratio of accurately predicted positive observations to the all predicted positives. A high precision value signifies that the model yields more relevant results compared to irrelevant ones. In this context, TP denotes True Positives, FP refers to False Positives, TN stands for True Negatives, and FN represents False Negatives.

$$\frac{TP}{TP + FP} \quad (14)$$

5.5.2. Recall

Recall (as shown in equation xv) is the ratio of accurately predicted positive observations to the total observations in the actual class. A high recall value indicates that the model successfully identifies most of the relevant results.

$$\frac{TP}{TP + FN} \quad (15)$$

5.5.3. F1 score

The F1 Score (as defined in equation xvi) represents the weighted average (Wavg) of Precision and Recall. It reflects the balance between these two metrics.

$$2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (16)$$

5.5.4. Specificity

Specificity (as outlined in equation xvii), also referred to as the true negative rate, quantifies the proportion of actual negatives that are accurately identified.

$$\frac{TN}{TN + FP} \quad (17)$$

5.5.5. Jaccard index

The Jaccard Index (equation xviii) assesses the similarity between sets of finite sample, and is used for evaluating the accuracy of classification algorithms.

$$\frac{TP}{TP + FP + FN} \quad (18)$$

5.5.6. Accuracy

Accuracy (as indicated in equation xix) is the ratio of accurately predicted observations to the total number of observations.

$$\frac{TP + TN}{TP + TN + FP + FN} \quad (19)$$

5.5.7. Matthews correlation coefficient (MCC)

MCC (as defined in equation xx) is utilized in machine learning to evaluate the quality of binary and multiclass classifications. It considers both true and false positives as well as true and false negatives.

$$\frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (20)$$

5.5.8. Kappa

Kappa (as described in equation xxi) is a statistical measure that assesses inter-rater agreement or reliability.

$$\frac{P_o + P_e}{1 - P_e} \quad (21)$$

P_o represents Observed Agreement (Accuracy), while P_e denotes Expected Agreement, which is calculated by summing the products of the row and column marginal probabilities in the confusion matrix and then dividing by the square of the total number of observations.

The ResViT FusionNet model exhibits strong overall performance in classifying diabetic retinopathy across five classes, with a high accuracy of 93.01% and solid metrics across all key measures (Table 2). Metrics like Precision, Recall, and F1 Score are calculated per-class. Metrics like Specificity, Jaccard, Accuracy, MCC, and Kappa summarize the model's overall performance, so they are computed across all classes

Table 2
Performance metrics of ResViT FusionNet model.

Metrics	Healthy/No DR	Mild DR	Moderate DR	Severe DR	Proliferative DR	ResViT FusionNet
Precision	0.992	0.909	0.864	0.892	0.825	0.9307
Recall	0.997	0.816	0.954	0.664	0.820	0.9300
F1 score	0.995	0.860	0.9070	0.731	0.823	0.9275
Specificity	–	–	–	–	–	0.9818
Jaccard	–	–	–	–	–	0.8749
Accuracy	–	–	–	–	–	0.9301
MCC	–	–	–	–	–	0.8944
Kappa	–	–	–	–	–	0.8935

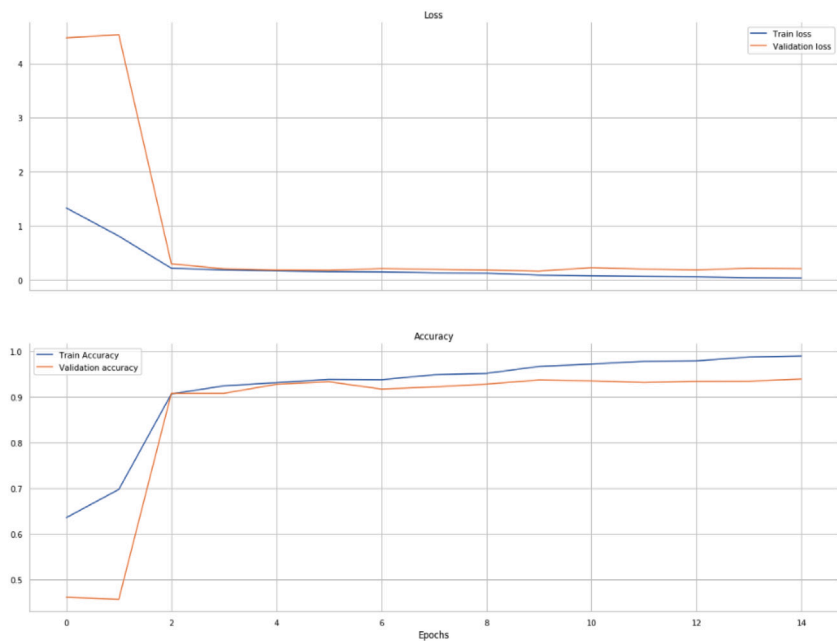


Fig. 8. Training and Validation Loss vs. Accuracy for the ResViT FusionNet model.

and filled only under the “ResViT FusionNet Model” column. Precision, recall, and F1 scores are consistently high, particularly for the Healthy class (precision: 0.992, recall: 0.997), indicating the model’s accuracy in identifying non-diseased cases. For Moderate DR, the model also performs well, with a precision of 0.864 and a recall of 0.954, suggesting effective detection of this category, which is crucial for early-stage intervention. The model shows slightly lower recall (0.664) for Severe DR, which, along with its F1 score of 0.731, indicates some challenges in detecting severe cases accurately. This may stem from overlapping features with neighboring stages, particularly Moderate DR and Proliferative DR. Despite these challenges, the overall specificity (0.9818), Jaccard index (0.8749), and high Matthews correlation coefficient (MCC) of 0.8944 reflect the model’s robustness and strong classification capability across all classes. Furthermore, a high Kappa score of 0.8935 indicates good inter-rater agreement, emphasizing the reliability of the ResViT FusionNet model. These metrics together highlight the model’s potential as a reliable tool for diabetic retinopathy detection, with its strengths in identifying early and moderate stages while leaving room for improvement in distinguishing more advanced stages.

During the early epochs (Fig. 8) the model learns simple features from the retinal images, leading to a rapid increase in accuracy. As training progresses, the model begins fine-tuning its feature representations to capture more complex patterns. This results in a slower,

more linear improvement in accuracy, reflecting the gradual convergence of the model to its optimal performance. The partially linear segment signifies this slower learning phase as gradients decrease and the model approaches saturation. The loss curve demonstrates a sharp decline in both training and validation loss during the initial epochs, indicating effective learning and good generalization. Similarly, the accuracy curves exhibit a rapid increase, eventually stabilizing around 99.1% (training) and 94.2% (validation). This consistency between training and validation metrics suggests that the model has effectively learned to distinguish between different DR stages without significant overfitting. The confusion matrix (Fig. 9) and ROC curves (Fig. 10) demonstrate that our diabetic retinopathy classification model achieves a solid performance, with a testing accuracy of 93.01%. The model performs exceptionally well in identifying No DR and Moderate DR cases, supported by the high true positive counts (1805 for No DR and 954 for Moderate DR) and strong ROC AUC values. Notably, No DR achieves an AUC of 1.00 due to its large sample size and clear distinction from other classes. The Moderate DR class also shows a high AUC of 0.95, benefiting from sufficient representation (999 samples). In contrast, Mild DR (AUC = 0.90) and Proliferative DR (AUC = 0.90) demonstrate reasonable accuracy, though they have fewer samples (370 and 291, respectively). These classes occasionally experience misclassifications, primarily into neighboring stages, as shown in Fig. 9.



Fig. 9. Confusion matrix of ResViT FusionNet model for Multilevel DR classification.

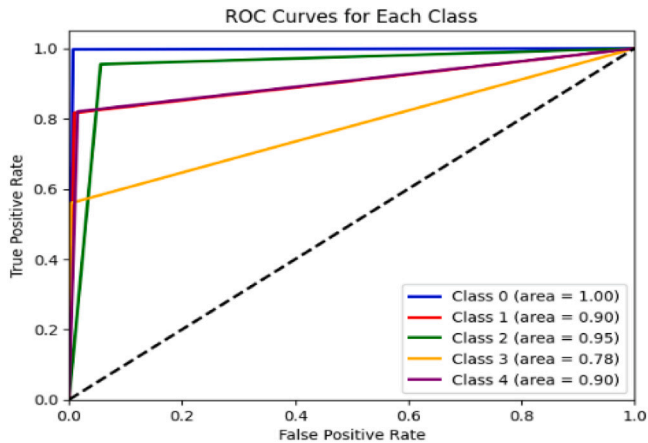


Fig. 10. ROC graph of ResViT FusionNet model for Multilevel DR classification.

For Severe DR, the model achieves a relatively lower AUC of 0.78, which aligns with the limited number of samples (192) and suggests challenges in distinguishing this class from adjacent stages (Moderate and Proliferative DR). This pattern indicates feature overlap between advanced stages, impacting classification performance. Overall, the model's strong performance in detecting early and moderate stages supports its potential as a reliable diagnostic tool. Future improvements should focus on enhancing separation between advanced stages by incorporating additional features, leveraging more balanced datasets, or exploring multi-modal imaging techniques.

5.6. Explainable AI (XAI) for interpreting diabetic retinopathy classifications

To improve the interpretability of our deep learning models for diabetic retinopathy (DR) classification, we incorporated two widely-used Explainable AI (XAI) techniques: Local Interpretable Model-agnostic Explanations (LIME) and Gradient-weighted Class Activation Mapping

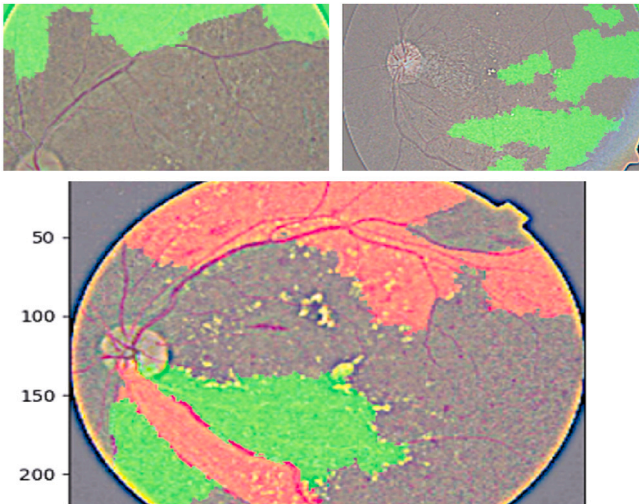


Fig. 11. LIME highlighting the influential parts of fundus images.

(Grad-CAM). These methods were crucial in bridging the gap between AI model predictions and clinical decision-making, which is essential for fostering trust and reliability in medical applications. Both techniques provide transparency into the model's decision-making process, which enhances its clinical adoption by allowing medical professionals to understand the basis of the model's predictions. LIME was applied to highlight the specific pixel-level regions in retinal images that significantly influenced the model's classification of different DR stages. The visualizations generated by LIME display red regions as areas of high influence, while green regions represent lower influence (Fig. 11). These fine-grained visualizations are crucial for understanding the model's behavior at a very detailed level. In practice, the LIME-generated heatmaps allow us to pinpoint specific retinal features, such as lesions, blood vessels, or microaneurysms, which are essential in distinguishing between DR stages. For instance, when classifying images as "Moderate DR", LIME highlights areas with microaneurysms

or small hemorrhages, aligning well with known clinical features of this stage of DR. This type of pixel-level explanation allows clinicians to validate the model's predictions by visually identifying specific regions of interest that the model considers important. This transparency builds confidence among medical practitioners, knowing that the model's reasoning aligns with established clinical knowledge.

In addition to LIME, Grad-CAM was used to generate broader, region-based heatmaps to visualize areas of higher significance for the model's decision-making. Grad-CAM works by computing the gradients of the predicted class with respect to the feature maps of the final convolutional layer, highlighting which regions of the image contributed most to the classification outcome. The generated heatmaps are superimposed on the original retinal images, with warmer colors (red and yellow) indicating regions with higher importance and cooler colors (blue) signifying lower relevance (Fig. 12). Grad-CAM provides a more general, holistic view compared to LIME, making it easier to identify larger patterns or abnormalities that influence the model's predictions. For instance, in the case of "Mild DR", Grad-CAM emphasizes localized regions corresponding to small microaneurysms, which are characteristic of early-stage DR. In contrast, for more advanced stages like "Severe DR" or "Proliferative DR", the heatmaps highlight larger areas of hemorrhages, exudates, and neovascularization, which are indicative of the more severe manifestations of the disease.

Together, LIME and Grad-CAM provide complementary forms of interpretability. While LIME allows for localized, pixel-level explanations by identifying specific regions that influence the model's decision, Grad-CAM focuses on global patterns, offering insights into the broader regions that the model is attending to. This combination not only enhances the interpretability of the model but also aligns it with clinical diagnostic workflows, helping clinicians understand the rationale behind the model's predictions in a way that is consistent with their clinical experience. For example, the pixel-level insights from LIME may guide clinicians to investigate certain areas of the retinal image more closely, while the broader heatmaps from Grad-CAM offer a quick overview of the larger abnormalities that are critical in grading the severity of DR. This dual approach significantly increases the clinical utility of the AI model, as it not only identifies the areas in the retinal image that are important for classification but also provides the reasoning behind the decision in a way that is accessible and understandable to clinicians. Furthermore, the combination of these XAI techniques ensures that the model is focusing on clinically relevant features of the retinal image, which supports the model's reliability as a decision-support tool. This interpretability is crucial for real-world applications of AI in medical settings, where transparency is key to ensuring the trust and adoption of AI-powered systems for diagnosing and grading diseases like diabetic retinopathy. Ultimately, the visualizations from LIME and Grad-CAM allow clinicians to validate AI-driven predictions, providing the necessary insights to make informed decisions in patient care.

5.7. Methodology for multilevel classification of DR

This algorithm (1) outlines the process for training and evaluating a hybrid ResNet50 and Vision Transformer (ViT) model for diabetic retinopathy classification using the APTOS 2019 dataset. It begins by loading pre-trained models and removing their classification layers, followed by combining features from both models. The dataset is preprocessed with pixel rescaling and data augmentation, and then split into training (80%), validation (20%), and test sets. The model is fine-tuned by unfreezing specific layers, trained over multiple epochs, and evaluated using metrics like precision, recall, F1-score, and accuracy. LIME is used for model interpretability and Grad-CAM is applied to visualize heat maps. The final trained model, confusion matrix, and explainability results are saved for future use.

Algorithm 1 Multilevel Classification of Diabetic Retinopathy (DR) using APTOS 2019 Dataset.

- 1: Load pre-trained ResNet50 and Vision Transformer (ViT) models, removing their classification layers.
- 2: Combine extracted features from ResNet50 and ViT using a fully connected layer.
- 3: Prepare the APTOS 2019 dataset by:
 - Rescaling the pixel values to a range of [0, 1].
 - Applying data augmentation techniques, such as horizontal flip, rotation, and zoom using ImageDataGenerator for training data.
- 4: Split the dataset into training, validation (80% training for training, 20% for validation), and test sets.
- 5: Initialize DataLoaders for the training, validation, and test datasets, set up CategoricalCrossentropyLoss, and configure the Adam optimizer with learning rate scheduling.
- 6: Fine-tune specific layers in ResNet50 and ViT by unfreezing them to improve model performance.
- 7: Train the model across multiple epochs, validate after each epoch, and apply early stopping if validation loss does not improve.
- 8: Save the optimal model according to validation loss and load it for the final evaluation.
- 9: Assess the model on the test dataset by calculating precision, recall, F1-score, accuracy, and other relevant metrics.
- 10: Generate the confusion matrix and apply Grad-CAM to visualize heatmaps with predicted probabilities, enhancing model interpretability. Additionally, apply LIME to further interpret the model's predictions.
- 11: Save the final trained model, confusion matrix, Grad-CAM heat maps, and explainability results for future use.

5.8. Ablation study of model configurations

To evaluate the individual contributions of various components in the proposed ResViT FusionNet model, an ablation study was conducted, and the results are summarized in the (Table 3). The baseline configuration using a CNN model (ResNet50) achieved an accuracy of 90.5%, with precision, recall, and F1-score values of 0.880, 0.885, and 0.882, respectively. These metrics provide a solid starting point, demonstrating the effectiveness of ResNet50 in extracting relevant features for diabetic retinopathy classification. However, the absence of any additional model enhancements or preprocessing techniques limits its performance.

In comparison, employing the Vision Transformer (ViT) as a standalone model yielded slightly lower performance metrics than the CNN baseline, with an accuracy of 89.8% and F1-score of 0.872. While ViT is effective in capturing global context and attention-based features, its reliance on large datasets and computational complexity likely constrained its performance when used in isolation. Introducing a hybrid architecture that combined ResNet50 and ViT without preprocessing yielded significant improvements, achieving an accuracy of 92.5% and an F1-score of 0.911. This hybrid approach leverages the complementary strengths of CNNs for local feature extraction and ViTs for capturing global dependencies. However, the absence of preprocessing still limited its potential.

The proposed ResViT FusionNet model, which incorporates advanced preprocessing techniques and a fusion strategy (fine-tuning), demonstrated the best overall performance, achieving an accuracy of 93.01%, precision of 0.9307, and an F1-score of 0.9275. These results highlight the cumulative benefits of preprocessing, hybrid modeling, and a carefully designed fusion strategy. The substantial improvement over intermediate configurations validates the importance of integrating preprocessing techniques to enhance the input quality and employing a fusion mechanism to synergize the strengths of CNNs and ViTs effectively. This ablation study underscores the progressive impact

Table 3
Ablation study results for different model configurations.

Configuration	Accuracy	Precision	Recall	F1-Score	Jaccard	MCC	Kappa
Baseline CNN (DenseNet201)	0.905	0.880	0.885	0.882	0.802	0.828	0.827
Vision transformer (ViT)	0.898	0.870	0.875	0.872	0.792	0.815	0.814
Hybrid CNN+ViT (No Preprocessing)	0.925	0.910	0.912	0.911	0.845	0.865	0.864
Proposed ResViT FusionNet model	0.9301	0.9307	0.9300	0.9275	0.8749	0.8944	0.8935

Table 4
Comparison of ResViT FusionNet model with DL models.

Models	Precision	Recall	F1-Score	Jaccard	MCC	Kappa
Proposed ResViT FusionNet Model	0.9307	0.9300	0.9275	0.8749	0.8944	0.8935
InceptionV3	0.7931	0.7741	0.7124	0.8154	0.8013	0.7532
EfficientNet50	0.8287	0.8041	0.7987	0.7861	0.8042	0.8041
DenseNet121	0.6521	0.6420	0.6015	0.6452	0.6543	0.6962

of each component, culminating in the superior performance of the proposed ResViT FusionNet model.

6. Comparative analysis

The results and interpretations based on the APTOS 2019 dataset are presented in this section, comparing the ResViT FusionNet model with other deep learning classifiers. Experiments were carried out on the APTOS 2019 dataset to assess the efficacy of these models (Table 4). While EfficientNet50 and InceptionV3 exhibit strong performance, they are less effective than the ResViT FusionNet model, with DenseNet121 delivering the least favorable results. The proposed ResViT FusionNet model outperforms EfficientNet50, InceptionV3, and DenseNet121 across all metrics in diabetic retinopathy classification. Its superior precision, recall, and overall alignment with true labels underscore its advanced architecture and suitability for this task.

We compare our ResViT FusionNet model with various ViT-based deep learning pipelines and other state-of-the-art models for identifying the severity stages of diabetic retinopathy utilizing retinal images obtained from fundus photography. As shown in Table 5, the ResViT FusionNet model significantly enhances performance in this task compared to other leading models, both CNN-based and ViT-based. The accuracy of ResViT FusionNet, 0.9301, surpasses several other advanced models, including ViT-based models such as ViT-CNN [53] (accuracy 0.825), the Swim Transformer [54] (accuracy 0.82), and the Discrete Wavelet Transform (DWT) CNN [52] (accuracy 0.7185). Additionally, the proposed model performs better than traditional CNN architectures like InceptionResNetV2 [55] (accuracy 0.5250) and CNN-based models [56] (accuracy 0.6668). In terms of F1-Score, ResViT FusionNet demonstrates its robustness with a score of 0.927, positioning it as one of the top performers in this comparison. Model such as Retinal ViT [57] (F1 0.919) also show strong performance, but ResViT FusionNet's ability to generalize better across all severity stages of diabetic retinopathy is evident. Notably, the model's high accuracy reflects its superior capacity for handling class imbalance, a common issue in diabetic retinopathy classification, as evidenced by its application of both CNN and transformer-based techniques. The results from this comprehensive comparison underline the effectiveness of the ResViT FusionNet model, which leverages both the spatial feature extraction power of CNNs and the contextual reasoning capabilities of Vision Transformers. The model's hybrid approach enables it to learn from both local and global image patterns, improving its ability to identify subtle lesions and features associated with different stages of diabetic retinopathy.

Table 5
Comparison of ResViT FusionNet with previously proposed models.

Models	Accuracy	DR classification level	Publication year	F1- Score
ViT model [58]	0.914	5 Classes	2021	–
InceptionRes-NetV2 [55]	0.5250	5 Classes	2021	–
CNN model [56]	0.6668	5 Classes	2022	–
ViT-CNN model [53]	0.825	5 Classes	2023	–
CNN model [59]	0.9167	5 Classes	2023	–
ViT model [10]	0.7254	5 Classes	2023	–
ViT + CSRA [60]	0.8235	5 Classes	2023	–
Swim transformer [54]	0.820	5 Classes	2023	–
DWT CNN [52]	0.7185	5 Classes	2024	–
RIQA Model [61]	0.877	5 Classes	2024	–
Retinal ViT [57]	–	5 Classes	2024	0.919
DWT swin transformer [62]	0.8600	5 Classes	2024	–
HFF-Net model [63]	0.7377	5 Classes	2024	–
Proposed ResViT FusionNet model	0.9301	5 Classes	–	0.927

7. Conclusion

In this paper, we present a new hybrid model called ResViT FusionNet, designed for the automated detection and classification of Diabetic Retinopathy (DR) by integrating Convolutional Neural Networks (CNNs) and Vision Transformers (ViT). The ResViT FusionNet leverages the strengths of both architectures: the robust feature extraction capabilities of ResNet50 and the long-range dependency capturing power of the Vision Transformer. This integration enables more accurate and reliable multilevel classification of DR, providing significant support to ophthalmologists for precise and comprehensive examinations. The proposed model's architecture, specifically designed to handle the complexities of retinal images, demonstrates superior performance across various metrics such as precision, recall, F1-score, and overall accuracy, when compared to other state-of-the-art deep learning models like EfficientNet50, InceptionV3, and DenseNet121. In addition to high accuracy, the model's interpretability is enhanced through the integration of Explainable AI (XAI) techniques, which provide valuable insights into the model's decision-making process. These techniques include LIME and Grad-CAM, offering visual explanations that highlight critical regions of the retina influencing the model's predictions. This transparency is essential for clinical validation, fostering trust among healthcare professionals and ensuring that the AI system aligns with medical expertise.

To ensure robust model training, we applied extensive data preprocessing and augmentation techniques. The ResViT FusionNet was fine-tuned to adapt high-level features effectively to the specific nuances of the DR dataset, improving both generalization and classification performance. The comprehensive evaluation demonstrates the model's effectiveness in diagnosing DR accurately. These results confirm that

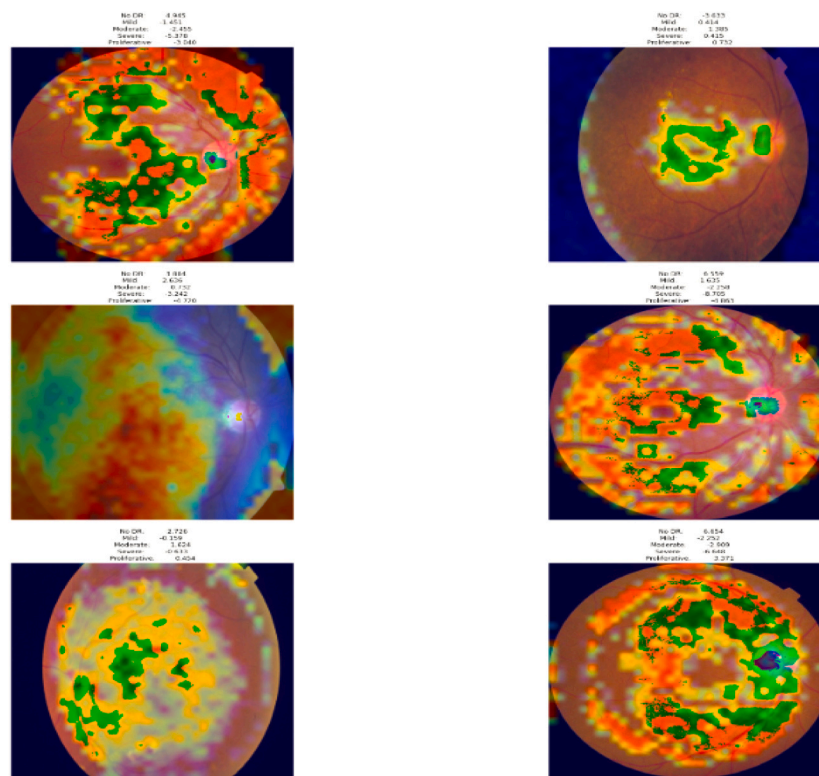


Fig. 12. Grad-CAM Heatmaps highlighting Model Attention for Diabetic Retinopathy Severity levels.

our proposed ResViT FusionNet model is a reliable and efficient tool for the initial identification and grading of Diabetic Retinopathy, advancing the field of medical image analysis and providing a valuable resource for ophthalmologists.

8. Future directions

Error analysis was conducted to identify common misclassification patterns in the proposed ResViT FusionNet model. A closer examination of the confusion matrix revealed that the majority of errors occurred between adjacent diabetic retinopathy (DR) severity levels, such as “Mild DR” being misclassified as “Moderate DR” or “Moderate DR” being mistaken for “Severe DR”. These errors can largely be attributed to the overlapping clinical features between these stages, such as the presence of microaneurysms and small hemorrhages, which may not be distinctly pronounced in fundus or OCT images. Furthermore, variability in image quality, illumination, and the presence of artifacts in the dataset also contributed to misclassifications. For instance, low-contrast images or images with occlusions, such as those caused by cataracts or inadequate pupil dilation, made it challenging for the model to distinguish subtle features.

To address these limitations in future work, several strategies can be implemented. First, augmenting the dataset with synthetic images using generative adversarial networks (GANs) or other data augmentation techniques can help balance the representation of DR stages. Second, employing more advanced preprocessing methods, such as contrast enhancement or artifact removal algorithms, can improve image quality and reduce noise. Third, incorporating multi-modal data, such as patient demographics, medical history, or OCT angiography (OCTA) images, could provide additional context for the model to better differentiate between severity levels. Leveraging ensemble methods or attention mechanisms could enable the model to focus more effectively on key regions of interest, further reducing errors. Finally, incorporating a continuous learning or federated learning approach might allow the model to be updated dynamically with new data from

various hospitals or medical centers, ensuring that it stays current with evolving data trends while maintaining privacy and security. These enhancements will likely improve the robustness and accuracy of the model, paving the way for more reliable clinical applications.

Integrating the ResViT FusionNet model into real-world clinical settings presents several challenges and limitations. One major concern is the significant computational resources required for training and deploying such a hybrid model, as it combines CNN and Vision Transformer architectures, which can be resource-intensive, particularly in under-resourced clinical environments. Additionally, potential biases in the dataset, such as imbalanced representation of different demographic groups or imaging conditions, could limit the model’s generalizability and fairness when applied to diverse populations. Validation across broader and more heterogeneous datasets, including images from various ethnicities, imaging devices, and healthcare settings, is essential to ensure robust and equitable performance. Addressing these challenges will require collaboration between AI researchers, clinicians, and healthcare institutions to optimize the model for real-world deployment while maintaining accuracy and fairness.

CRedit authorship contribution statement

Amna Ikram: Writing – original draft, Methodology, Conceptualization. **Azhar Imran:** Writing – review & editing, Supervision.

Ethics

This research was conducted in accordance with the highest ethical standards and principles. The study does not involve human participants, animal subjects, or data from identifiable individuals, and hence, no ethical approval was required. All data used in this study were obtained from publicly available sources or generated through computational methods. Proper citations and acknowledgments have been provided for any external datasets, algorithms, or methodologies used.

The authors affirm that this work is original, has not been published previously, and is not under consideration for publication elsewhere. Furthermore, all authors have contributed significantly to this work and have approved the final manuscript for submission. There are no conflicts of interest or competing interests to disclose. Any funding sources that supported this research are acknowledged in the manuscript. The authors are committed to maintaining transparency, integrity, and adherence to ethical guidelines throughout the research and publication process.

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Materials availability

N/A.

Code availability

It is available on request from corresponding author.

Declaration of competing interest

None.

Data availability

The dataset used for this study is publicly available at the following link: <https://www.kaggle.com/datasets/mariaherrero/aptos2019> [APTOS 2019 Dataset].

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